

THEO HATZIS

Wideband Mixed-Signal / Power / Photonics & Specialist Electronics *Post-Silicon & Semiconductor Test Equipment | Physical Systems, Sensors, MedTech, SciTech, Electrochemistry, Batteries & Electronics | MeasX*

Location: UK & DACH Ecosystem (Munich Hub) **Engagement:** Senior Fractional Specialist / SOW / Contract **Contact:** theohatzis@gmail.com | +44 (0) 7869 507965 | [Theo Hatzis](#) | [LinkedIn](#)

EXECUTIVE SUMMARY

Senior fractional specialist across mixed-signal post-silicon, power-device test (GaN/SiC), analogue front-ends, photonics/optical instrumentation, battery BMS/EIS electronics, scientific/medical instrumentation, sensors, and electrochemical/battery R&D. Operates across the full lifecycle: design-input, low-parasitic DUT/testboard layout, functional verification, measurement-integrity validation, and system-level debug. Delivers into Osram, Broadcom, Infineon, Renesas, TI, Advantest-type semiconductor programmes, as well as MedTech/SciTech, physical-systems, and outstation-engineering environments involving semiconductor test equipment, WBG static/limits/DPT, analogue/mixed-signal ICs, optical/VCSEL/LED systems, sensors, and battery-system electronics.

TARGET PROFILE & ENGAGEMENT SCOPE

- **Target Clients & Ecosystem:** Osram • Broadcom • Infineon • Renesas • Texas Instruments • Munich/DACH Semiconductor Ecosystem • UK Science & Technology • Medical Technology & Diagnostics • Highways/Infrastructure (Outstation Engineering)
- **Core Domains:** Mixed-signal post-silicon • Power devices (WBG GaN/SiC) • Photonics • Battery systems & BMS/EIS • Semiconductor test equipment • Electrochemistry R&D • Scientific/medical instrumentation • Sensors & physical-systems engineering • Outstation electronics (Highways)
- **Role Focus:** Senior / Principal Engineer – Mixed-Signal, Power, Photonics, Post-Silicon Validation, Scientific/Medical Instrumentation, Battery/Electrochemical Systems
- **Engagement Mode:** Fractional specialist • Project-defined technical scope • Design-input + functional verification • Cross-domain physical-systems engineering

PROFESSIONAL SPECIALISM

Cross-domain specialist spanning mixed-signal electronics, power-device test, photonics/optical instrumentation, battery BMS/EIS and electrochemical systems, scientific/medical instrumentation, sensors, and semiconductor test-equipment development. Operating across the complete post-silicon lifecycle — from design-input and low-parasitic DUT/testboard layout to functional verification, measurement-integrity validation, and system-level debug.

Leverages a deep technical background in chemical, electrochemical, spectroscopic, and physical-systems methods to link surface chemistry, material degradation, and sensor/measurement-chain behaviour directly to electrical performance. Supports semiconductor, MedTech/SciTech, battery-system, and physical-systems clients building WBG GaN/SiC test platforms, analogue/mixed-signal ICs, power modules, optical/photonics devices, sensors, and battery-system electronics. Can support advanced battery and electrochemical chipsets including EIS (Electrochemical Impedance Spectroscopy).

CORE TECHNICAL COMPETENCIES

Semiconductor Test Equipment — Design & Functional Verification

- Design-input for DUT testboard, switching cells, and measurement chains

- Low-parasitic PCB layout for high-speed switching and high-bandwidth sensing
- Functional verification of static, limits, and dynamic DPT test modules
- Mixed-signal stability under high dv/dt, high di/dt, EMI, and thermal stress
- Validation of converter stages, gate-drive behaviour, and protection mechanisms
- Measurement-integrity analysis: artefact identification, bandwidth constraints, transient distortion

WBG GaN/SiC Device Testing

- Static I-V, leakage, RDS(on), and threshold stability
- SOA / FBSOA / RBSOA limits testing
- Dynamic DPT switching: Eon/Eoff, reverse recovery, transient behaviour
- Gate-drive optimisation under extreme switching conditions
- Platform verification competing against Keysight B1500 / PD1500A-class systems

Analogue & Mixed-Signal Post-Silicon

- ATE / PXI / TestStand automation pipelines
- Mixed-signal measurement physics and sensor-chain behaviour (optical, inertial, chemical)
- ADC/DAC linearity, timing, synchronisation
- Reliability drift, long-cycle characterisation, cross-domain debug
- Functional verification of analogue front-ends (AFEs) and mixed-signal ICs

Photonics / Optical / Sensing Instrumentation

- Optical / IR / UV acquisition pipelines
- Photonics test setups (LED / laser / VCSEL, Osram/Broadcom-aligned; diagnostic optics)
- High-speed optical measurement electronics
- Mixed-signal integration with photonics front-ends

Scientific / Medical Instrumentation & Diagnostics (MedTech / SciTech)

- Spectroscopic pipelines (Raman, IR, UV-Vis, fluorescence) for analytical/diagnostic systems
- NMR / XRF / EDXRF acquisition chains and detector-front-end behaviour
- Optical/chemical sensor integration for medical/scientific platforms
- Mixed-signal stability in diagnostic and analytical instrumentation
- Physical-systems debug across optics, sensors, and mixed-signal AFEs

Battery BMS & EIS Mixed-Signal Post-Silicon

- EIS / impedance spectroscopy electronics
- DRT-ready acquisition and mixed-signal stability
- BMS analogue front-end (AFE) validation
- Relaxation signatures and calendar-aging interpretation
- High-precision measurement chains for battery analytics

Electrochemistry & Battery R&D

- Electrode/interface kinetics and degradation modelling
- EIS/DRT interpretation pipelines for cell and module behaviour
- Electrochemical sensor integration and mixed-signal acquisition
- PV → battery → inverter autonomous system validation

- Correlation of electrochemical signatures with electrical performance

Physical Systems & Sensor-Chain Engineering

- Pressure, inertial, chemical, and optical sensor-chain validation
- Mixed-signal integration with physical-systems measurement pipelines
- Reliability, drift, and environmental stress behaviour in sensor systems
- Cross-domain debug: sensor → AFE → mixed-signal → ATE
- Outstation electronics validation for highways/infrastructure sensing systems

Software-Defined Validation, AI Pipelines & Build Infrastructure

Test Automation & Executives: Object-Oriented Python (OOP) framework built on PyTest (fixtures, parametrization) and NI TestStand sequences for automated DUT characterisation. Instrument control via IVI-COM, PyVISA, SCPI, and C#/.NET drivers.

Applied AI & ML Analytics: Python vectoring & RAG pipelines for automated datasheet/doc query synthesis. TensorFlow (DNN/DLN) models for waveform pattern recognition and anomaly detection. Proprietary MephistoDB for AI-driven feature extraction and long-cycle drift analysis.

Reporting, Data & Modeling: Automated HTML/PDF verification logging via Jinja2 and Office Interop; desktop GUIs in C#/.NET / PyQt. Parametric time-series parsing in Pandas with interactive 2D/3D Plotly visuals. Transfer-function & signal modeling in MATLAB.

Build Systems & IDE Tooling: Multi-language build orchestration using CMake and Ninja. Environment management via Conda/Virtualenv across Visual Studio (.NET/C#), PyCharm Professional, and VS Code environments.

Semiconductor Metrology & Physical Measurements

Front-End / Back-End / Materials Metrology

- SEM / STEM pipelines for defectivity, morphology, cross-sectional analysis
- AMO measurement chains for thin-film and interface behaviour
- XPS for surface chemistry, oxidation states, contamination analysis
- XRF for elemental composition, film thickness, dopant profiling
- Raman / IR / UV-Vis spectroscopies for lattice, phonon, band-structure signatures
- Application of metrology pipelines to scientific/medical instrumentation
- Correlation of physical-measurement artefacts with electrical post-silicon behaviour

2D/3D Materials, Dielectrics, Sensors & MEMS

- Characterisation of 2D materials (graphene, TMDs) for mobility, interface states, contact behaviour
- 3D materials and dielectrics: permittivity, leakage, breakdown, reliability drift
- Sensor-chain metrology: pressure, inertial, chemical, optical sensors
- MEMS structural/electrical co-validation: resonance, damping, stiction, packaging effects
- Integration of materials-metrology outputs into mixed-signal and ATE pipelines

Physical Measurements + Electrical PVT Integration

- Linking physical-measurement artefacts directly to electrical PVT signatures
- Cross-domain debug: Metrology → Analogue Front-End → Mixed-Signal → ATE
- Reliability, drift, ageing, environmental stress correlations
- Measurement-integrity validation across electrical and physical domains

SUMMARY OF TECHNICAL FOCUS AREAS

- High-power electronics • converter design • gate-drive systems
- Mixed-signal stability in disturbed environments (high dv/dt, high di/dt)
- WBG GaN/SiC switching physics • photonics/optical measurement electronics (VCSEL / LED / laser)
- Battery BMS/EIS mixed-signal pipelines • electrochemical R&D • PV→battery→inverter systems
- Scientific/medical instrumentation • sensors • diagnostic optics
- ATE / PXI / TestStand automation • software-defined validation pipelines
- Measurement physics • reliability • long-cycle characterisation
- Low-parasitic PCB layout • high-bandwidth sensing
- Physical-measurement pipelines (SEM / STEM / XPS / XRF / spectroscopies)
- Materials / sensors / MEMS metrology • physical-systems integration

SENIOR PROFILE & ENGAGEMENT MODEL

- **Profile:** Senior / Principal Engineer with broad semiconductor experience across power, mixed-signal, photonics, battery-system electronics, and scientific/medical instrumentation. Strong background in post-silicon validation, semiconductor test-equipment development, electrochemical/EIS systems, sensors, and physical-systems engineering. Acts as a seamless technical interface between internal R&D teams and external development partners. Hands-on engineering mindset paired with structured documentation and clear execution. UK + DACH delivery arc with direct exposure to Osram, Broadcom, Infineon, Renesas, TI-type semiconductor programmes, as well as MedTech/SciTech and physical-systems environments.
- **Engagement Terms:** Fractional Specialist • SOW-Based Delivery • Project-Defined Scope / Payload • Mixed-Signal / Power / Photonics Post-Silicon Validation • Semiconductor Test-Equipment Development • Scientific/Medical Instrumentation • Battery/Electrochemical Systems • Contract / 3-Month Rolling FTC Welcome (Outside IR35 / B2B)

EXTENDED TECHNICAL LOG & LONGFORM CV

For comprehensive project histories, schematics scope, and extended metrology case studies:

- **Download Longform CV (PDF):** [Theo_Hatzis_prf.pdf](#)
- **Technical Grid:** [TechnicalCapabilityGrid](#)